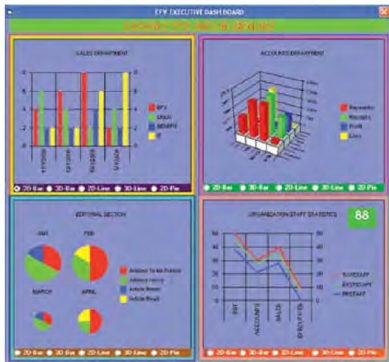




Screenshot 2: Program output of executive dashboard with arbitrary data

User can include them while designing for specific application.

Executive dashboard development

Form design. Run Visual Basic and open a new project and form. Position the form appropriately on the screen, resize the controls and set their properties as shown in the table.

Normally, ADO and MSChart controls are not available in the general control toolbox. To add these controls to the project, go to project → components and check the boxes against Microsoft ADO Data Control 6.0 (OLEDB) and Microsoft Chart Control 6.0 (OLEDB) options. The symbols of these two controls appear in the general control toolbox on the left-hand side of the screen. Place the controls on the form. By default, the SeriesColumn property of chart control is set to '1'. Don't change this value, as it selects the first column for the plot in X-axis.

Program coding

The program is simple and self-explanatory. Here, it is designed to display sales department, accounts department, editorial section and the organisation's staff statistics. There are

four database tables: dash 1, dash 2, dash 3 and dash 4 containing the records for sales, accounts, editorial and staff statistics respectively. All these tables are in 'efyexecutive.mdb' file created in MS Access application. The MSChart controls can be connected to the database tables using the code in the subrou-

line `init_chart()` given below:

```
Private Sub INIT_CHART()
On Error Resume Next
' INITIALIZE SALES CHART
Set MSChart1(3).DataSource = Nothing
Adodcl1.RecordSource = "DASH1"
Adodcl1.Refresh
Set MSChart1(0).DataSource = Adodcl1
MSChart1(0).Stacking = False

'INITIALIZE ACCOUNTS CHART
Set MSChart1(0).DataSource = Nothing
Adodcl1.RecordSource = "DASH2"
Adodcl1.Refresh
updatefields 1, 2, 3
Set MSChart1(1).DataSource = Adodcl1
MSChart1(1).Stacking = False

'INITIALIZE EDITORIAL CHART
Set MSChart1(1).DataSource = Nothing
Adodcl1.RecordSource = "DASH3"
Adodcl1.Refresh
updatefields 1, 2, 3
Set MSChart1(2).DataSource = Adodcl1
MSChart1(2).Stacking = False

'INITIALIZE ORGANIZATION CHART
Set MSChart1(2).DataSource = Nothing
Adodcl1.RecordSource = "DASH4"
Adodcl1.Refresh
Set MSChart1(3).DataSource = Adodcl1
MSChart1(3).Stacking = False
```

PRSTAFF_UPDATE

End Sub

A few fields in tables 'dash2' and 'dash3' require calculation. These are calculated in the subroutine `updatefields()`:

```
Private Sub updatefields(p As Integer, q As Integer, r As Integer)
Dim k As Integer
While Adodcl1.Recordset.EOF = False
k = (Adodcl1.Recordset.Fields(p) . Value - Adodcl1.Recordset.Fields(q) . Value)
If Sgn(k) = -1 Then
Adodcl1.Recordset.Fields(r).Value = Abs(k)
Adodcl1.Recordset.Fields(r + 1). Value = 0
Else
Adodcl1.Recordset.Fields(r).Value = 0
Adodcl1.Recordset.Fields(r + 1).Value = Abs(k)
End If
Adodcl1.Recordset.Update
Adodcl1.Recordset.MoveNext
Wend
Adodcl1.Refresh
End Sub

The 'prstaff' field in 'dash4' table is incremented to reach the total existing staff, then it is decremented until it reaches zero. Thus the graph corresponding to prstaff moves up and down between totstaff and exstaff graphs and updated every second to show the effect of real-time processing. The real-time updated data of the present staff (prstaff) is indicated in the textbox control using the prstaff_update() subroutine:

Private Sub PRSTAFF_UPDATE()
Adodcl1.Refresh
Select Case MaxBool
Case False:
While Adodcl1.Recordset.EOF = False
If Adodcl1.Recordset.Fields(3) . Value < Adodcl1.Recordset.Fields(2) . Value Then
Adodcl1.Recordset.Fields(3).Value = Adodcl1.Recordset.Fields(3).Value + 1
M = M + 1
Text2.Text = M
End If
Adodcl1.Recordset.Update
```